

Multiproposal Elliptical Slice Sampling

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³UCLA

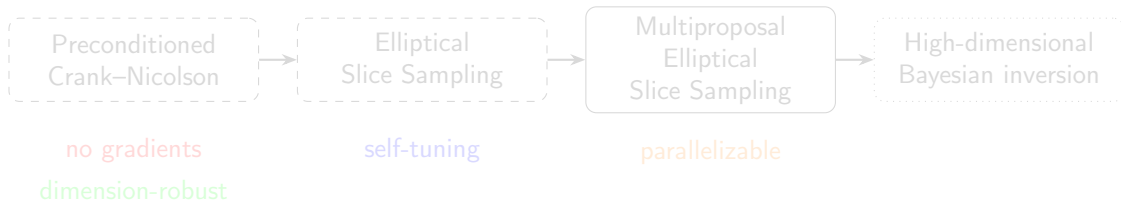
PDE seminar at Department of Mathematics, Drexel University

Philadelphia, 2026 June 5

Summary

Goal: Scalable MCMC for high-dimensional posteriors with form

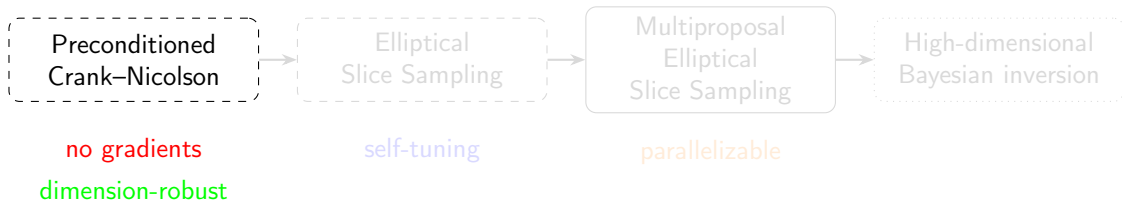
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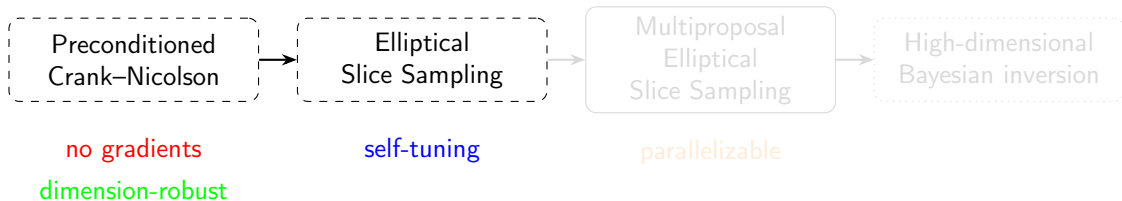
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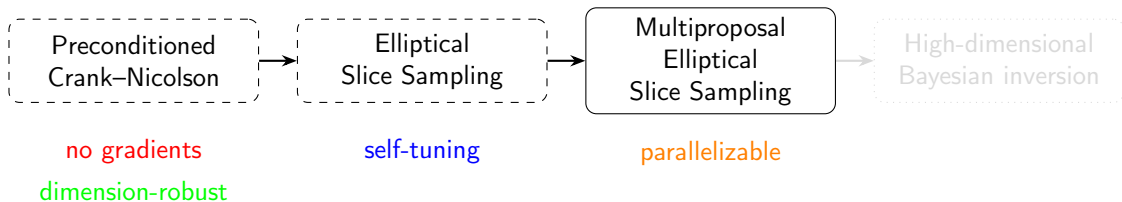
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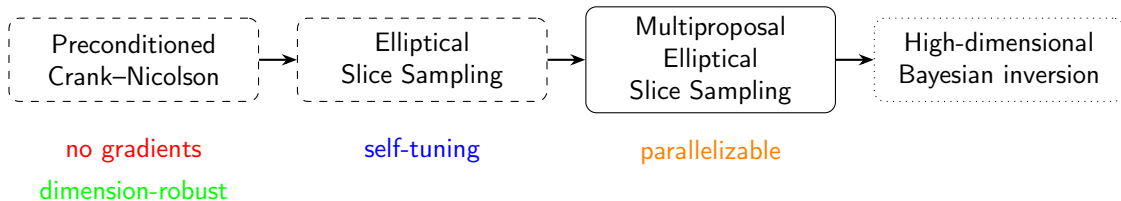
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Motivation

- Bayesian inverse problem

$$\pi(x|y) \propto \mathcal{L}(y|x) \pi_0(x), \quad y \in \mathbb{R}^k \text{ (observations)}, \quad x \in \mathbb{R}^d \text{ (parameters)}$$

- Often non-parametric:

$$\{x(s) : s \in D\}, \quad D \subset \mathbb{R}^d.$$

- Sampling with Markov models:

- Construct ergodic Markov chain $X = \{X_1, X_2, \dots, X_N\}$ with π as invariant measure.
- Approximate expectations:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N f(X_i) = \hat{\mathbb{E}}_{\pi}[f(X)]$$

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Markov chain Monte Carlo (MCMC)

Algorithm 1: Metropolis-Hastings (Hastings, 1970)

Input: Current state x , likelihood $\mathcal{L}(y|x)$, prior $\pi_0(x)$, proposal density q .

1. Sample $x^* = x + \nu$, $\nu \sim q$
2. Compute acceptance probability

$$\alpha(x, x^*) = \min\left\{1, \frac{\mathcal{L}(y|x^*)}{\mathcal{L}(y|x)} \times \frac{\pi(x^*)}{\pi(x)} \times \frac{q(x|x^*)}{q(x^*|x)}\right\}$$

3. Accept x^* with probability $\alpha(x, x^*)$, otherwise reject.
-

$$\text{total cost}_\epsilon = \text{cost per iteration} \times \text{number of iterations}_\epsilon$$

- Both increase with input size.

Pre-conditioned Crank-Nicolson (PCN)

Posterior distribution¹:

$$\pi(x|y) \propto \mathcal{L}(y|x) \mathcal{N}(x; 0, C).$$

Algorithm 2: PCN (Cotter et al., 2013; Neal, 1999)

Input: Current state x , aggressiveness $\rho \in (0, 1)$

1. Propose $x^* = \rho x + \sqrt{1 - \rho^2} \nu$, $\nu \sim \mathcal{N}(0, C)$ no gradients
 2. Compute $\alpha(x, x^*) = \min\{1, \mathcal{L}(y|x^*)/\mathcal{L}(y|x)\}$
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Proposal is reversible with respect to the prior:

$$\mathcal{N}(x^*; 0, C) \mathcal{N}(x; \rho x^*, (1 - \rho^2)C) = \mathcal{N}(x; 0, C) \mathcal{N}(x^*; \rho x, (1 - \rho^2)C) \Rightarrow \text{dimension-free}$$

¹Stuart (2010); Dashti and Stuart (2013)

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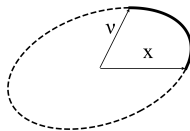
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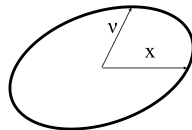
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Elliptical Slice Sampling (ESS)²



(a) PCN: $x^* = \rho x + \sqrt{1 - \rho^2} \nu$, $\rho \in (0, 1)$



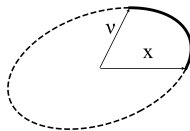
(b) ESS proposal

- Elliptical proposal:

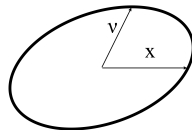
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- Holds the prior invariant.
- Choose ρ automatically $\rightarrow$ self-tuning.

Elliptical Slice Sampling (ESS)²



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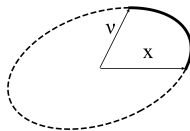
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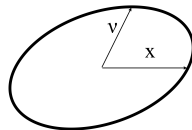
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²Murray et al. (2010)

Slice Sampling

- Idea:

$$\pi(x, y) = \begin{cases} \frac{1}{\int f(x) dx} & 0 < y < f(x) \\ 0 & \text{otherwise} \end{cases}$$

Algorithm 2: Slice sampling (Neal, 2003)

1. Sample $y \sim \mathcal{U}(0, f(x))$
 2. Sample $x \sim \mathcal{U}(\mathcal{S})$
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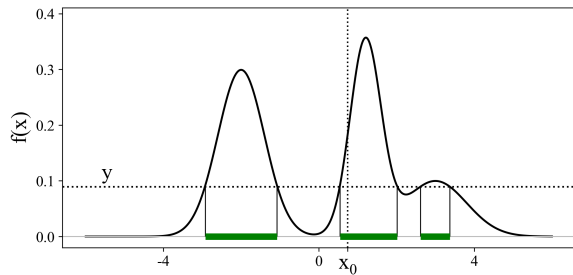


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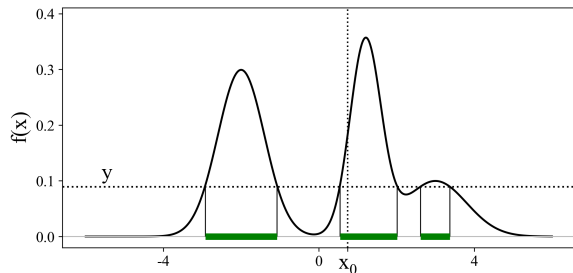


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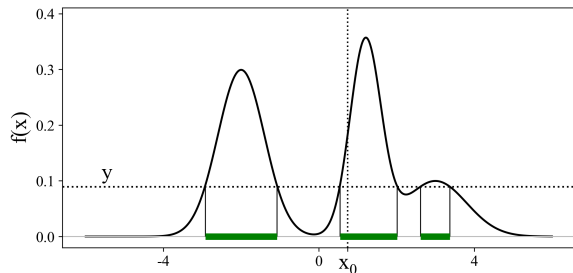
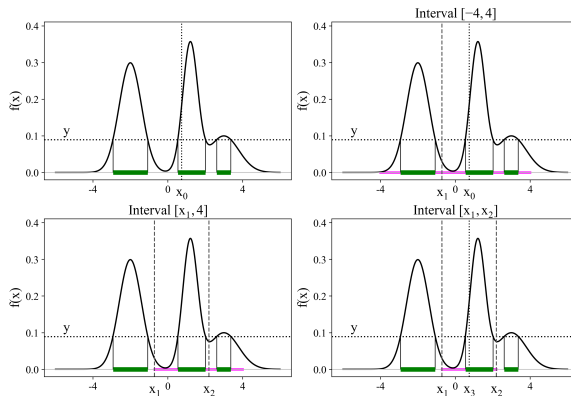


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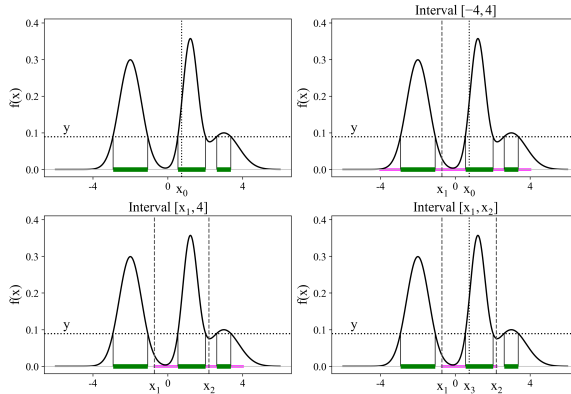
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- How to sample x on $\mathcal{S}$ when $\mathcal{S}$ is unknown?
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Elliptical Slice Sampling (continued)

Algorithm 4: ESS

Input: Current state x .

Draw $\nu \sim \mathcal{N}(0, C)$, defining the ellipse

Draw $\theta_x \sim \mathcal{U}(0, 2\pi)$

Draw slice level $y \sim \mathcal{U}(0, \mathcal{L}(y|x))$

Set $\theta_{\min} \leftarrow 0$, $\theta_{\max} \leftarrow 2\pi$

while $\mathcal{L}(y|x^*) < y$ **do**

 Sample $\theta \sim \mathcal{U}(\theta_{\min}, \theta_{\max})$

 Propose $x^* = x \cos \theta + \nu \sin \theta$

 Evaluate $\mathcal{L}(y|x^*)$

 Shrink $(\theta_{\min}, \theta_{\max})$ around θ_x

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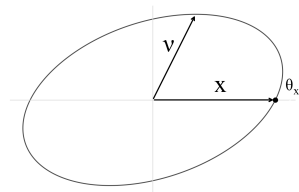
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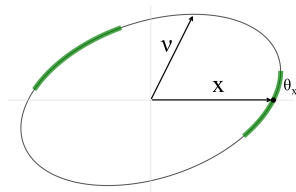
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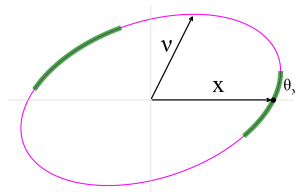
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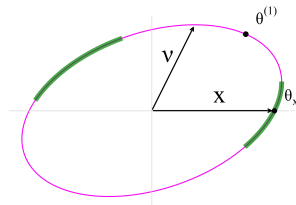
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One iteration of Elliptical Slice Sampling

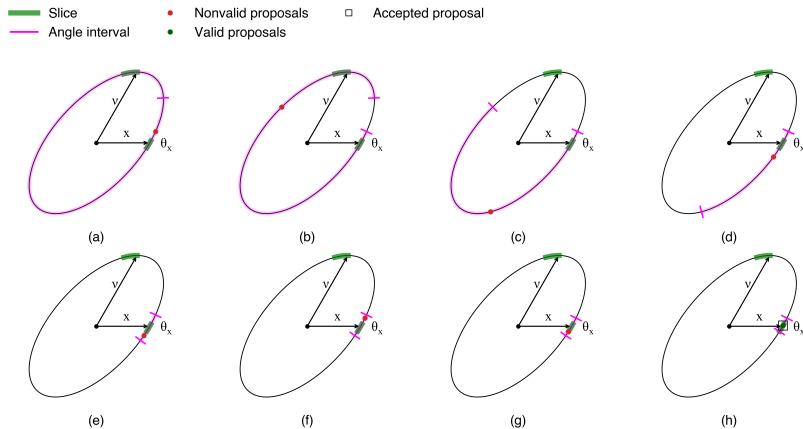


Figure: One iteration of ESS.

Multiproposal Elliptical Slice Sampling

Algorithm 5: MESS (single iteration)

Input: Current state x , number of proposals M .

Draw $\nu \sim \mathcal{N}(0, C)$, $\theta_x \sim \mathcal{U}(0, 2\pi)$, $y \sim \mathcal{U}(0, \mathcal{L}(y|x))$

Set $\theta_{\min} \leftarrow 0$, $\theta_{\max} \leftarrow 2\pi$

while $|\mathcal{A}| = 0$ **do**

 Draw $\theta^{(j)} \sim \mathcal{U}(\theta_{\min}, \theta_{\max})$, $j = 1, \dots, M$ **parallelizable**

 Propose $x^{(j)} = x \cos \theta^{(j)} + \nu \sin \theta^{(j)}$

 Compute $\mathcal{A} = \{j : \mathcal{L}(y|x^{(j)}) \geq y\}$

 Shrink using all angles

Select $m \in \mathcal{A}$, set $x \leftarrow x^{(m)}$.

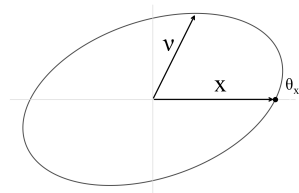


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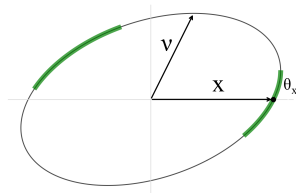


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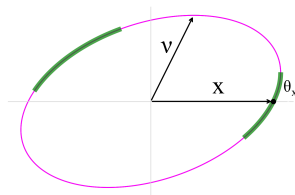


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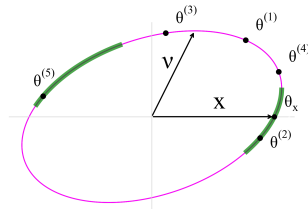


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Mess in action

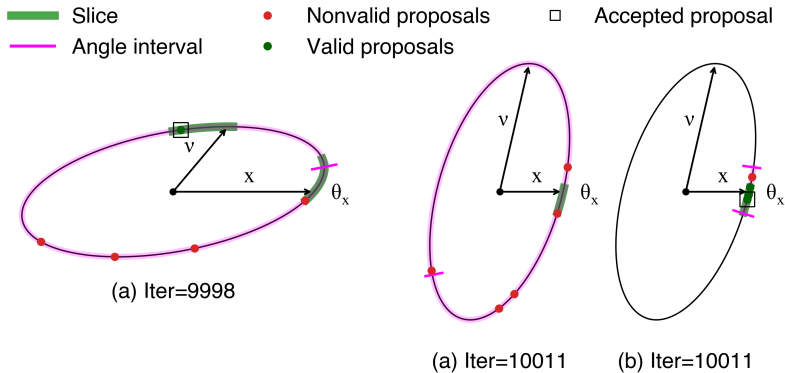


Figure: Two iterations of MESS with $M = 5$.

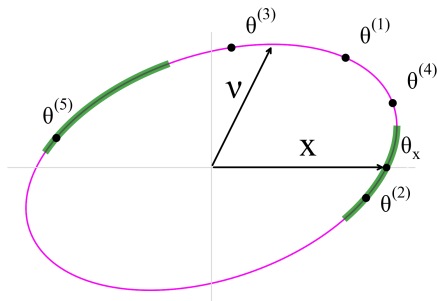
Acceptance step with transition matrix

- Sample $\theta^{(1)}, \dots, \theta^{(5)}$.
- Valid indices: $\mathcal{A} := \{j : \mathcal{L}(x^{(j)}) \geq y\} = \{2, 5\}$.
- Transition matrix with entries

$$P = \begin{bmatrix} P_{22} & P_{2x} & P_{25} \\ P_{x2} & P_{xx} & P_{x5} \\ P_{52} & P_{5x} & P_{55} \end{bmatrix}$$

- Naive choice (uniform):

$$P^{\text{unif}} = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$



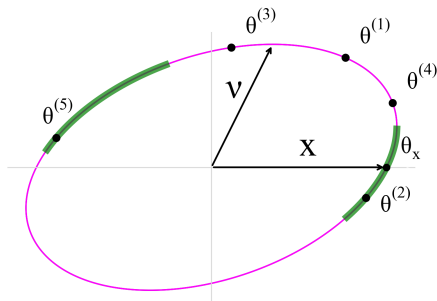
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- Naive choice (uniform):

$$P^{\text{unif}} = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$



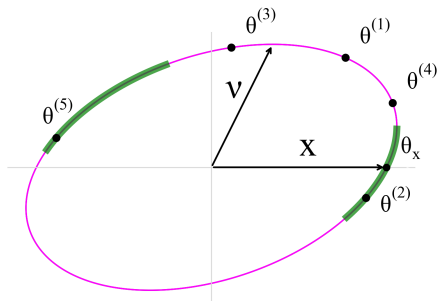
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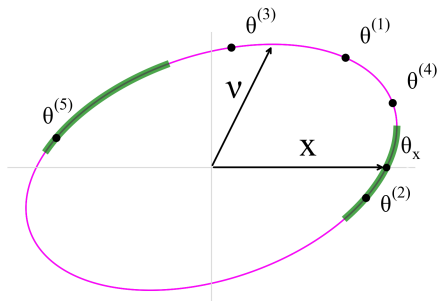
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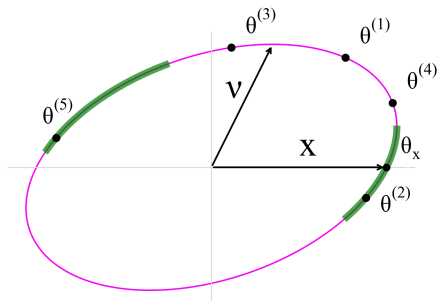
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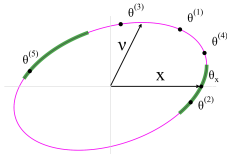


Distance-informed transition matrix

- Use distance to bias transitions:

$$\max_{\mathbf{P}} \sum_{r=1}^3 \sum_{s=1}^3 d(r, s) P_{rs}$$

subject to double-stochastic $\mathbf{P}$ with zero diagonal.



- Angular distance matrix:

$$D_{ang} = \begin{bmatrix} 0 & 0.3 & 3 \\ 0.3 & 0 & 2.7 \\ 3 & 2.7 & 0 \end{bmatrix} \quad (\text{rad})$$

- Example of optimized matrix:

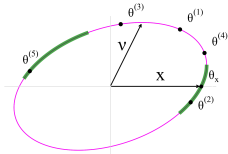
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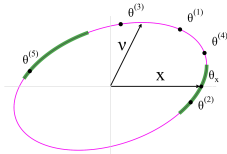
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Mixing improvement

- Uniform MESS.
- M cores.

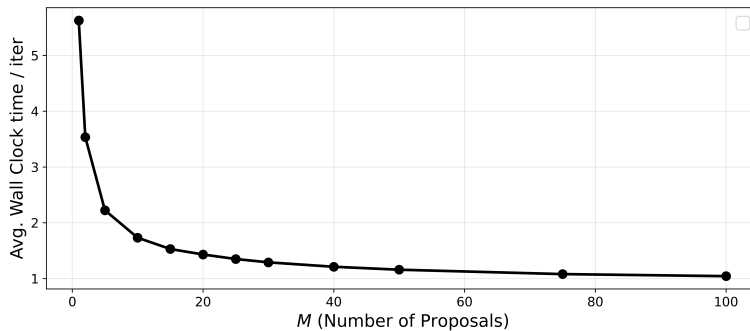


Figure: Average number of subiterations (for a logistic regression model).

Solute transport model³

- Mimics transportation of a solute by a fluid under damping and external forcing.

$$\frac{ds}{dt} + (\mathbf{A} + \kappa \mathbf{I}_d) \mathbf{s} = \mathbf{g}.$$

- $\mathbf{s} \in \mathbb{R}^d$: Concentration.
- $\mathbf{g} = (1, 0, \dots, 0) \in \mathbb{R}^d$: Forcing.
- $\kappa \in \mathbb{R}$: Damping.
- $\mathbf{A} \in \mathbb{R}^{d \times d}$: Spectral transport matrix.
- Steady state:

$$\mathbf{s}(\mathbf{A}) = (\mathbf{A} + \kappa \mathbf{I}_d)^{-1} \mathbf{g}.$$



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Bayesian inverse problem

- Gaussian prior on the non-zero elements of $\mathbf{A}$:

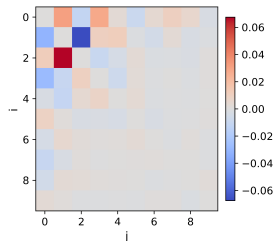
$$\mathbf{A} \sim N(0, \mathbf{C}), \quad \mathbf{C} = \text{diag}(\{\text{Var}(a_{ij})\}).$$

- Zero-diagonal.
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- $m = d(d-1)/2$ elements in upper triangle;
- $\mathbf{C}$ penalizes long range interactions and high frequencies.
- Gaussian likelihood:

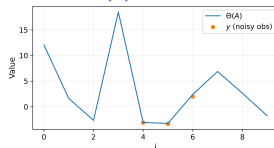
$$y = \mathbf{s}(\mathbf{A}) + \eta, \quad \eta \sim N(0, \sigma^2 \mathbf{I}_d).$$

- Posterior:

$$\pi(\mathbf{A}|y) \propto \mathcal{L}(y|\mathbf{A})\pi_0(\mathbf{A}).$$



(a) $\mathbf{A}_{10}$.



(b) $\mathbf{s}(\mathbf{A}_{10})$ and y_3 .

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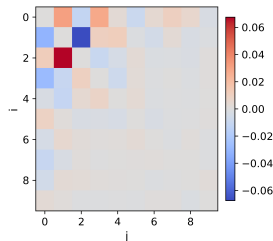
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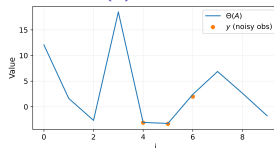
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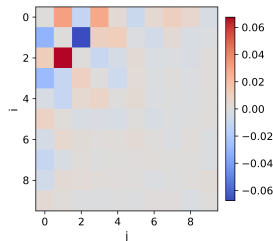
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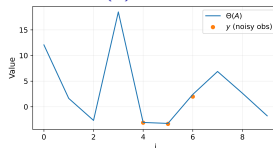
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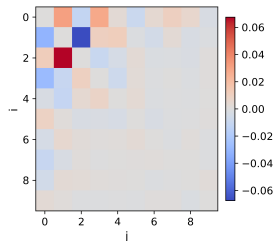
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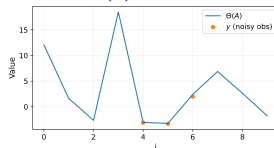
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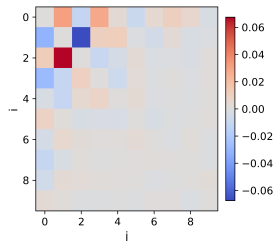
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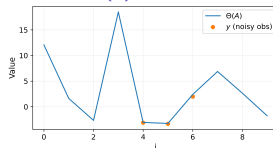
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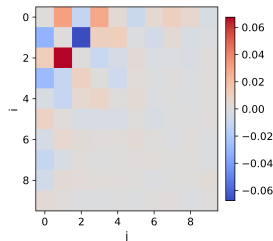
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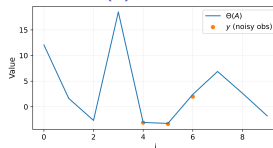
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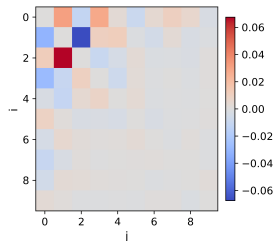
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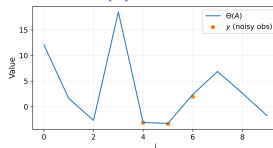
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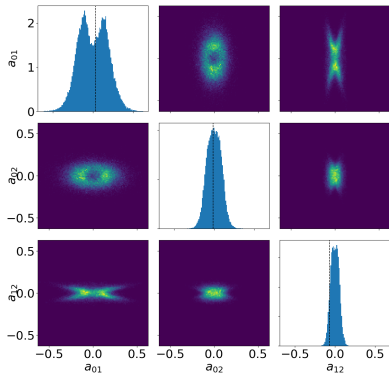


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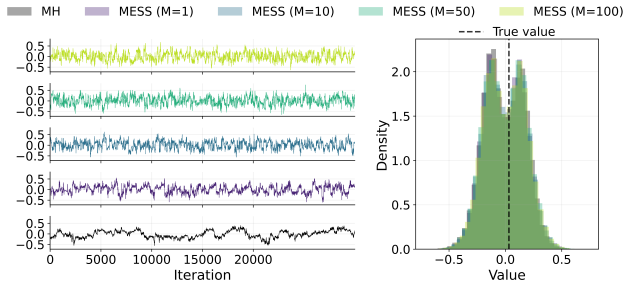


(b) $\mathbf{s}(\mathbf{A}_{10})$ and y_3 .

Posterior distribution for $d = 10$



(a) Some marginals and pairwise densities.



(b) Chain realizations for a_{01} .

Dimension-free behavior

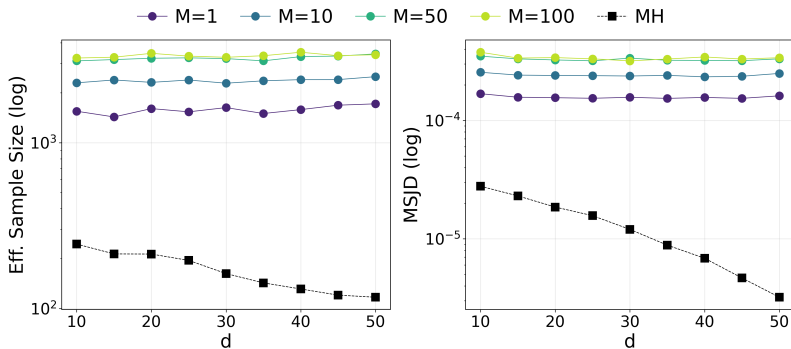


Figure: MESS mixing rate does not degrade with dimension.

Conclusions

Multiproposal Elliptical Slice Sampling (MESS) is

Self-tuning

no gradients

dimension-robust

parallelizable.

Useful for large-scale and/or non-parametric Bayesian estimation with Gaussian priors.



References I

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Comparison with Gibbs and HMC: blind deconvolution inverse problem

- Observation model:

$$d = w \star c + e, \quad w \sim N(0, C_w), \quad c \sim N(0, C_c), \quad e \sim N(0, C_e).$$

- Posterior is multimodal (sign-shift symmetry).

